
Research Proposal

MSc Thesis in Data Science and Business Analytics

[Optimization of exam schedule for International Baccalaureate]

by

[Tan Yujing and Student number is 13319027]

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Introduction

The International Baccalaureate (IB) examination timetabling problem is a complex large-scale scheduling problem in which a set of exams must be assigned to limited time slots while satisfying multiple institutional constraints. These constraints include restrictions on exam clashes, the sequencing of exams and the broader objective of spreading exams across the timetable in a way that reduces student burden. Due to the large scale and possible subject combinations of the IB, creating a high-quality schedule is not only important in practice but also computationally challenging.

Furlong (2024) found that the optimization of the IB examination schedule can be expressed as a mixed-integer linear optimization (MILP) model. Although this method is theoretically attractive as it offers a structured way to represent rules and optimize the quality of the schedule, the results on real world big size data show that precise optimization is computationally infeasible. Although feasible solutions can be found, solving the optimality of large-scale instances requires a considerable amount of computing time. This has created a gap between what can be solved in theory and what can actually be applied in practice.

This research proposes to examine heuristic methods as a more implementable alternative. The goal of heuristic methods is to generate high-quality schedules within a realistic time frame, rather than seeking optimal solutions. This method can iteratively improve a feasible schedule under the guidance of a penalty-based evaluation function through operations such as exchanging exams or applying other local modifications.

Research questions

How to design and evaluate heuristic methods to effectively generate high-quality IB exam schedules for large-scale datasets

Methodology and Techniques

Local search heuristics / Matheuristics

Data and/or expected results

For this research, I expect to work with data related to the IB examination timetabling problem. A relevant dataset may become available, subject to confirmation that it can be shared. If this is not possible, synthetic data may be provided as an alternative.

The expected outcome of this study is not necessarily a verified optimal schedule, but rather than develop one or more heuristic methods that can generate high-quality feasible schedules within a reasonable computing time. The research will focus on identifying methods that perform well in practice, evaluating the quality of their solutions, and examining their computational efficiency. Meanwhile, where possible, compare the results of the newly proposed heuristic solution with those of the existing MIP optimization model.

Literature [Key References]

Furlong, A. (2024) '*Optimisation of the International Baccalaureate Diploma Programme Examination Timetable*'. Amsterdam Business School.

E. S. K. Siew, S. N. Sze, S. L. Goh, G. Kendall, N. R. Sabar and S. Abdullah. (2024) 'A Survey of Solution Methodologies for Exam Timetabling Problems.' *IEEE Access*, vol. 12, pp. 41479–41498, 2024.
<https://ieeexplore.ieee.org/document/10473052>

R. Qu, E. K. Burke, B. McCollum, L. T. G. Merlot & S. Y. Lee. (2009). 'A survey of search methodologies and automated system development for examination timetabling.' *Journal of Scheduling* 12, 55–89 (2009).
<https://link.springer.com/article/10.1007/s10951-008-0077-5>

Timetable

1st Month (April)

Activity	Date
1, I will formulate a detailed research plan and conduct a key literature review on the heuristic algorithms and related scheduling methods for the examination schedule. 2, I will familiar and test with the existing datasets, pension-based evaluation functions, and any code snippets provided by my supervisor. 3, In addition, I will also set up the coding environment and implement an initial benchmark heuristic algorithm.	Weekly meetings (of 30 minutes)

2nd Month (May)

Activity	Date
1, I will develop and test several heuristic algorithms for the IB exam schedule problem. 2, Comparing the feasibility, solution quality and running time of different algorithms. 3, Meanwhile, I will start writing the main chapters of the thesis, especially the introduction, theoretical background, literature review, data, methodology and experimental design sections.	Weekly meetings (of 30 minutes)

3rd Month (June)

Activity	Date
1, I will complete the experiment, analyze the results and finish the first draft of the paper. 2, After receiving the feedback, I will make revisions and finalize the draft. 3, Preparing the final presentation and practice oral presentations.	Weekly meetings (of 30 minutes)

I am enrolled in the course:

Master's Thesis Data Science

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Master's Thesis Business Analytics

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Supervisor

Joaquim Gromicho

Signature



Second marker (suggestion by supervisor)

Dick den Hertog

Thesis coordinator

C.G.H. Diks

Signature